Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/30/2006
PI Name:	Watson, Andrew Ph.D.		
Project Title:	Human Centered Visual Displays		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Ames Research Center		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-01
Start Date:	10/01/2000	**End Date:**	09/30/2004
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Vision is the dominant input channel to the human brain, and the human brain is a critical component of every NASA mission. Visual information is thus central to the success of NASA missions. This information may be employed for operational, scientific, biomedical, psychosocial, administrative, historical, or public relations purposes. The quality of this information can determine whether docking maneuvers are hazardous, whether scientific discoveries are made, and whether missions return safely to earth. We propose to develop a new method for evaluating the quality of visual information, and with aid of these results, to provide tools for automatic evaluation of visual quality and principled design of human-centered visual displays. We propose to measure visual quality using new techniques that will provide an absolute, rather than relative measure, in units of JND (just-noticeable-difference). These units should be portable across contexts, labs , types of visual material, and applications. It should be possible to produce calibrated samples of specific quality ratings for test, specification, and demonstration purposes. Our initial research will be on digital video,		

	since this emerging as the preeminent medium of visual communication in advanced networks and application. Digital video is also an important application for quality evaluation, since it is usually compressed, and the degree of compression, and hence quality, has enormous impact on costs, bandwidth, safety, and ease of use. This work will be conducted in cooperation with national and international standards bodies that are eager to establish standards for subjective and objective evaluation of quality in digital video. Results of this research will be used to evaluate and improve our patent-pending technology for automatic evaluation of digital video (DVQ).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Earth benefits of this research will be manifest in any enterprise that relies on digital communication of visual information. Significant examples are medical imaging, Earth resource imaging, space imaging, science imaging, and internet imaging. In each case, there is a need for efficient archiving and distribution of digital images, and high quality display of those images. Advances in medical imaging in particular may be expected to enhance diagnostic capabilities and to reduce costs of medical care. Earth resource imaging may be expected to reduce environmental damage and reduce costs of detection and repair of such damage.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)